

Freight Rate Prediction

Machine Learning Engineer Assessment

Development data	48,000 labeled loads
Final prediction data	12,000 unlabeled loads
Final model	CatBoost with log-transformed target
Temporal CV MAE	\$107.15
Temporal CV RMSE	\$626.96
Temporal CV R ²	0.8271

This report documents the data exploration, quality checks, validation strategy, feature experiments, model selection, error analysis, final prediction pipeline, and the required December 2025 scenario forecast.

The assessment instructions require a PDF/DOCX report describing the train/test validation and split approach and including the fixed December prediction chart. The 12,000-row validation set is an unlabeled prediction set; therefore, no local ground-truth score is claimed for those final predictions.

Submission status

- 12,000 validation predictions generated and validated.
- 31 December scenario predictions generated and validated.
- Official scorer completed successfully and generated the required chart.

1. Objective

The objective was to build a machine learning system that predicts the posted freight rate for each load using the labeled development dataset provided with the assessment. The resulting model was then applied to every row in the 12,000-load validation dataset to produce the required submission file.

The development data contains shipment attributes such as origin, destination, geographic coordinates, supplied distance, equipment type, weight, date, market index, quote signal, and the target posted_rate. The final validation dataset contains the same load-level predictors but does not expose the target.

Assessment-aligned deliverables

- A reproducible modeling pipeline with Python dependencies and run instructions.
- validation_predictions.csv containing exactly load_id, predicted_rate.
- A PDF report documenting the validation/split methodology.
- The fixed December 2025 prediction chart generated by the provided scorer.

2. Data Understanding

The labeled development dataset contains 48,000 rows and 14 columns, including the posted_rate target. The final validation dataset contains 12,000 rows and 13 predictor columns. The development data covers January through October 2025, while the validation loads begin in November 2025.

Dataset	Rows	Columns	Target
Development	48,000	14	posted_rate
Validation	12,000	13	Not provided

Target distribution

The posted_rate target is right-skewed. Its mean is \$2,373.98, median is \$2,030.76, standard deviation is \$1,486.49, and maximum is \$25,533.00. The 99th percentile is \$5,972.83. This large upper tail was an important consideration when selecting the training objective.

Key relationships

Distance was by far the strongest simple numerical relationship with the target: its Pearson correlation with posted_rate was 0.9085. Weight, market_index, and quote_signal had much weaker linear correlations, while the geographic coordinates provided additional location information.

3. Exploratory Data Analysis and Data Quality

Missing values

Column	Development	Validation	Treatment
weight	300 (0.625%)	165 (1.375%)	Invalid/negative values treated as missing
market_index	374 (0.779%)	249 (2.075%)	Left missing for model handling

Additional quality checks

- No duplicate rows or duplicate load_id values were found in either dataset.
- No zero or negative distances were found.
- 292 negative weights were found in development data and 145 in validation data.

- No zero weights were found; negative weights were treated as invalid and converted to missing.
- No negative or zero target rates were found in the development data.

Geographic distance investigation

A Haversine distance was calculated as an exploratory feature. The supplied distance and Haversine distance were extremely strongly correlated (0.9995). The supplied distance averaged 1,135.86 miles versus 966.14 miles for Haversine distance. Because the supplied distance already captured the dominant signal and the feature experiments did not improve generalization, the final model retained the original distance representation.

4. Validation Strategy

Arandomtrain/testsplit was avoided because freight rates are time-dependent and the assessment validation data occurs after the labeled development period. Instead, an expanding-window temporal validation strategy was used.

Four monthly validation folds were constructed. Each fold predicts a future month using all available labeled data from earlier months. This better approximates the real deployment condition in which the model must predict future loads from historical observations.

Fold	Training rows	Validation rows	MAE	RMSE	R ²
July	28,806	4,912	\$111.25	\$627.26	0.8262
August	33,718	4,759	\$99.21	\$615.09	0.8261
September	38,477	4,670	\$110.62	\$619.04	0.8349
October	43,147	4,853	\$107.51	\$646.44	0.8211
Average	—	—	\$107.15	\$626.9	0.827

MAE was used as the primary business-facing metric because it directly expresses the average absolute pricing error in dollars. RMSE was retained as a secondary metric because it is sensitive to large errors, which is useful for monitoring expensive misses. R² was used as a general measure of explained variance.

Why temporal validation matters

The fold results are relatively stable in R² (0.8211–0.8349), while October has the highest RMSE at \$646.44. This suggests that large-rate observations remain difficult even when the model explains most of the overall variance. The temporal design also avoids leaking future information into earlier validation folds.

5. Feature Engineering and Ablation

Several additional features were investigated, including Haversine distance, route identifiers, latitude/longitude differences, weight-per-mile, missing-value indicators, calendar features, and cyclic representations of month and weekday.

Ablation results

Feature set	MAE	RMSE	R ²
Original features	\$112.44	\$629.50	0.8257
Engineered features	\$132.76	\$629.81	0.8255
Engineered + distance difference	\$139.40	\$630.97	0.8248

The ablation study showed that adding the broader engineered feature set did not improve the temporal validation score. In particular, the engineered feature set increased average MAE from \$112.44 to \$132.76. The distance-difference feature was also removed after it further degraded MAE and R^2 . This supported keeping the final model relatively simple rather than adding features solely because they were available.

The final model uses the original predictive fields plus a numeric date representation used by the final prediction pipeline. The target transformation, rather than a large expansion of the feature space, produced the most useful improvement.

6. Model Selection

CatBoost was selected as the primary model family because the dataset contains a mixture of numeric and categorical shipment attributes, nonlinear relationships, and a relatively small-to-medium tabular dataset. CatBoost also provides strong performance without requiring a large neural network pipeline.

Normal-target experiments

Multiple CatBoost configurations were evaluated using the same temporal folds. The best normal-target experiment achieved an average MAE of approximately \$126.91, with RMSE around \$628.49 and R^2 around 0.8262.

Log-target experiment

Because `posted_rate` is strongly right-skewed and contains a long upper tail, the target was log-transformed for training. Predictions were converted back to the original dollar scale for evaluation and submission. The log-target model reduced average MAE to approximately \$107.15 while maintaining similar RMSE and slightly improving R^2 .

Model	Avg. MAE	Avg. RMSE	Avg. R^2
Best normal-target CatBoost	\$126.91	\$628.49	0.8262
Final CatBoost + log target	\$107.15	\$626.96	0.8271

Final selection: CatBoost with the log-transformed `posted_rate` target. The final model was retrained on all 48,000 development rows before generating the 12,000 validation predictions.

7. Error Analysis

Error analysis was performed to understand where the selected model struggled. The most important pattern was that absolute error increased with shipment distance.

Distance bucket	Count	MAE	Median absolute error
<500 miles	1,079	\$45.97	\$16.24
500–1,000	1,520	\$71.11	\$27.69
1,000–1,500	890	\$120.85	\$44.12
1,500–2,000	595	\$162.18	\$64.83
2,000–3,000	711	\$212.95	\$74.39
>3,000	58	\$148.30	\$107.53

Equipment-level errors were comparatively similar: Dry Van MAE was \$100.43, Flatbed MAE was \$106.04, and Reefer MAE was \$124.99 in the evaluated error-analysis sample. This indicates that the dominant challenge was not simply equipment category but the combination of route length and unusual rate observations.

Largest-error observations

The worst individual errors were very high-rate loads where the observed posted_rate was far above the model prediction. For example, several loads had actual rates above \$14,000 while predictions were in the approximately \$2,000–\$6,000 range. These observations explain why RMSE remains much higher than MAE and indicate a sparse high-rate regime that is difficult to learn from the available features.

This error analysis supports the decision to use log-target training: it reduces the influence of extreme target values on the overall fitting process while improving the typical absolute prediction error.

8. Final Prediction Pipeline

After modelselection, thefinal CatBoost log-targetmodel was retrained on all 48,000 labeled development rows. It was then applied to the 12,000 rows in validation.csv.

Submission validation

- 12,000 prediction rows generated.
- Exactly two submission columns: load_id and predicted_rate.
- No duplicate or missing load IDs.
- All validation IDs aligned with the expected TE-000001 through TE-012000 set.
- No non-positive predicted rates.

The completed prediction file was then checked by the supplied scoring utility. The scorer reported: 'Validated 12,000 final predictions.'

9. December 2025 Scenario

The assessment provides a fixedDecemberscenario in which the shipment characteristics remain constant and only the date changes. The scenario is Lexington to Fort Wayne, 360 miles, Dry Van, 32,000 lb, with one row for each day from December 1 through December 31, 2025.

The final model produced positive predictions for all 31 days. The predicted rates range from \$828.96 to \$841.23, with a mean of approximately \$832.85. The relatively narrow range is consistent with the fixed route, equipment, distance, and weight while date-related inputs vary.

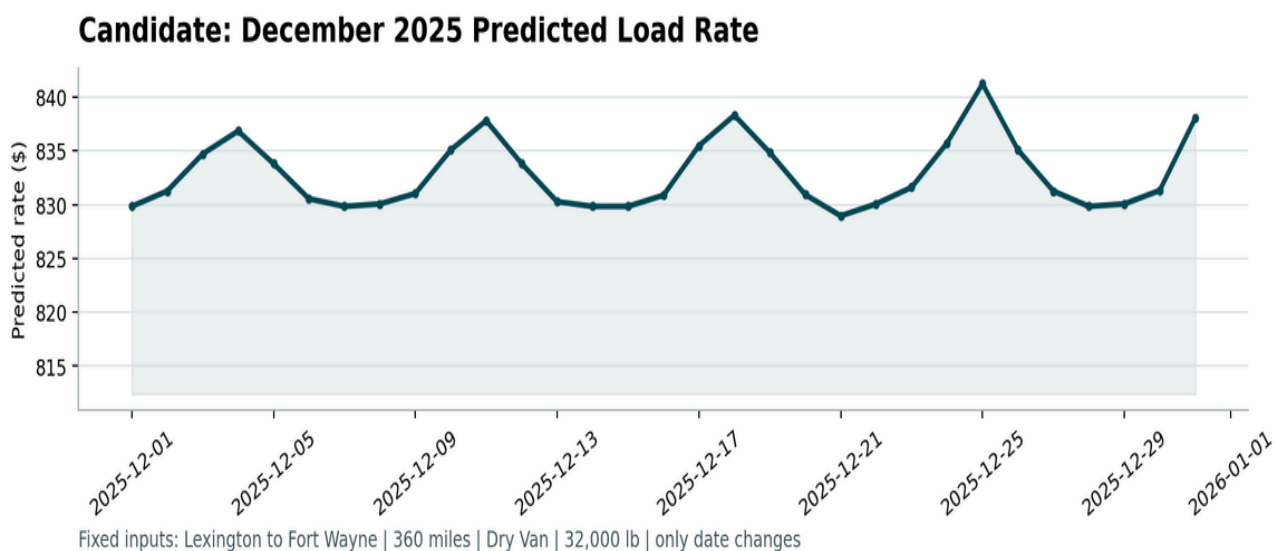


Figure 1. Fixed December 2025 prediction chart generated by the provided assessment scorer.

The official scorer validated all 31 December rows and generated the chart successfully. The scorer also states that final validation metrics for the 12,000-row submission are calculated by Spotter after submission.

10. Reproducibility and Project Structure

The solution is organized into separate modules for exploration, feature experimentation, model training, prediction, error analysis, and December forecasting. Dependencies are listed in requirements.txt.

```
src/  
eda.py  
features.py  
model_experiments.py  
error_analysis.py  
train.py  
predict.py  
predict_december.py  
inspect_december.py  
  
outputs/models/  
catboost_final_log_target.cbm  
  
scorer_results/  
candidate_december.png  
  
validation_predictions.csv  
requirements.txt  
README.md
```

The assessment datasets are supplied by the hiring team. The repository is intended to contain the solution code, dependencies, run instructions, final prediction file, model artifact, and required report/chart rather than republishing the supplied assessment data.

11. Conclusion

The final solution uses a temporally validated CatBoost regression model with a log-transformed target. The key modeling decision was to prioritize realistic future-month validation and robust handling of the target's long upper tail rather than aggressively increasing feature complexity.

The final temporal cross-validation results were MAE \$107.15, RMSE \$626.96, and R^2 0.8271. The model was retrained on the complete 48,000-row development dataset and successfully generated the required 12,000 validation predictions. The supplied scorer also validated the required 31-row December scenario and generated the final prediction chart.

The main remaining limitation is the sparse high-rate tail: large individual pricing errors remain concentrated among unusual high-rate loads. Future improvements could focus on richer route-level market features, external freight-market signals, and specialized treatment of rare high-rate regimes, subject to the data available in a production setting.

Final model: CatBoost regression with log-transformed posted_rate